

Economic Benefits Redistribution Methodology for Renewable Energy Communities

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Abstract

A large number of renewable energy communities are expected to be created in the near future, and, also in view of the funding that has been allocated in the various national climate and energy plans, there will be interest in their development from entities with different needs belonging to the institutional, industrial, commercial, and tertiary sectors. Given the possibility that these entities will be participated in by a variety of parties (individuals, SMEs, local governments, etc.) and in view of the particular "ethical nature" of RECs, it is to be expected that the remuneration of members will be in accordance with a principle of equity and in proportion to the ability to generate income for the community itself. For these reasons, a categorization of members according to their characteristics as renewable energy producers, prosumers or mere consumers is necessary. In this context, this article, with reference to an example case related to current legislation in Italy, proposes a model for the distribution of benefits related to energy production and sharing within the renewable energy community. These benefits concern those of the energy produced and simultaneously shared/absorbed by the members of the energy community as well as those related to the valorization of the energy fed into the grid, in excess of the consumption of the users of the energy community itself. The results obtained by applying the proposed methodology for the redistribution of the economic benefits accrued by the community among its participants show the usefulness of this distribution model in quantitative and qualitative terms, according to principles of equity and proportionality.

1 Introduction

Energy communities, first outlined in the European directives Renewables Directive (RED II) [1], "Clean Energy Package for all Europeans" [2] and in the Electricity Market Directive (IEM) [3], and recently implemented in the Italian regulatory framework, envisage the introduction of new players in the future energy market where active and passive users can cooperate in energy production and its local optimal sharing and management [4]-[7]. Renewable energy communities emerge as innovative entities able to aggregate different active and passive users who can thus cooperate and being involved in the development of energy production and its local optimal sharing and management. This revolution is the objective of the application the European and national policies to promote and raise awareness of the final consumer, whether private citizen or legal subject, towards a rational and thrifty use of energy to achieve the EU objectives of climate neutrality as well as pursuing the goals regarding the ecological and energy transition defined in the national resiliency plan.

Energy communities loom large in the energy transition program outlined by the European community as a key player given their ability to attract private sector investment and ability to contribute to the process of public acceptance of energy projects aimed at harnessing renewable resources. In addition, the benefits for the communities affect multiple areas and are not only limited to alleviate electricity bills, but also aim to mitigate pollution and revitalize local economies through the creation of new jobs. Citizens can thus contribute actively and more responsibly to the energy transition.

In Italy, RECs directive [1] has recently been transposed in the national regulatory framework [8][9][10] and are currently in an introductory and experimental phase and RECs have the opportunity to offer themselves as a solution to the market and consumers to curb rising energy prices. This is also why RECs require a careful preliminary study phase to achieve an optimal design [11][12]. During this phase, it is necessary to identify the ideal sizing of the community, which depends on the number of potential members, their attitude to consumption, existing renewable production facilities and potential new installations. The objective is to try to find an optimal combination to maximize the economic and environmental benefits also in consideration of the financial constraints. RECs fit into the Italian legal framework as a subject of law and ethical and fairness principles must guide the operations of the established body. The entity will have to equip itself with a statute and, most likely, internal regulations to manage the relations between the participating members. At this point, it is also necessary to define a policy for redistributing the amounts of payments received by the community body from the sale of energy or from state subsidies. A detailed energy-sharing mechanisms for energy community members has been proposed in [13] where redistribution of profits among community members are based on different ownership and sharing strategies. In this paper, different classes of users will be introduced, and each will be the recipient of part of the returns earned by the REC. The model that will be defined is based on principles of proportionality and meritocracy, i.e., the users can directly determine the percentage of benefit that can

be perceived by adopting virtuous behaviour but, at the same time, a remuneration base can be guaranteed that should incentivise small consumers to join the community effort. The redistribution of revenues among users will be performed by means of a top-down de-compositional approach of the total amount that has to be distributed. The model can be interpreted as a tree diagram in which the depth and number of leaves can be determined as required; therefore, many variations can be formulated. The proposed approach has as its strengths in its modularity and ease of adaptation to a generic context. This work is intended as a first step in the development of a more complex system and aims to demonstrate the validity of the chosen approach. A description of the approach is therefore provided and then a mathematical model is formulated. Finally, a case study is presented that shows the feasibility and usefulness of the methodology developed in this paper.

2. Methodology

2.1 Revenue redistribution approach

The incentive for creating RECs is typically based on the awarding of an economic benefit when energy is produced and consumed among users of the energy communities. In this paper the approach proposed for REC revenue sharing among members is based on the de-composition of total revenues into categories. Each category has associated a certain amount to be redistributed among users proportionally to the energy fed into or withdrawn from the grid, depending on the user class. A class of users may or may not compete in a category, depending on its class. Relatively to the revenue structure, are considered the inflows deriving from the shared energy incentive and the revenues from the “RID” (Dedicated collection, “Ritiro dedicato”) which consists of the remuneration by the GSE of electricity fed into the grid by plants that can access it, at the request of the producer and as an alternative to the free market, according to principles of procedural simplicity and applying market economic conditions. In this context it has been decided to perform only the redistribution of the sharing incentive, while the RID accrued by the single member is directly assigned to the correspondent user. This choice is due to the fact that those benefiting from the RID have incurred an investment for the construction of the plant, so they require an higher income to return on that investment. If, on the other hand, the plant is owned by the community, the valorisation of the energy deriving from the RID may be redistributed differently. In this case will be illustrated the redistribution of the shared energy incentive. In a similar way, the redistribution of the RID-derived amounts for community-owned plants may be realised. Before introducing the redistribution tree, which will be used to map the categories, it is required to define the class of the members of the energy community:

- Producer: a user equipped with renewable generation plants which directly feeds energy into the grid, like a PV plant connected to the network. The self-consumption is generally only that at the inverter's stand-by operation during the hours of absent insolation; therefore, its withdrawals from the grid are very low, almost negligible.

- Prosumer: this kind of member of the energy community are, for example, public, residential, commercial, or industrial users with photovoltaic systems. These are users that have the ability to self-consume part of the energy produced but may feed surplus energy into the power grid.
- Consumer: a user who does not have the ability to feed energy into the grid, but only to make withdrawals for the electric utilities it uses during its operation.

The user classes receive a percentage of the amount allocated to the categories proportionally to the energy withdrawal or injection realised in the various months. For the generic user are then defined:

$$w_{i,j} = \frac{E_{i,j}^w}{E_j^w}, \quad (1)$$

Where $E_{i,j}^w$ is the energy withdrawn by utility i during the j -th period and E_j^w is the total energy withdrawn from the grid by the community during the period j .

$$i_{i,j} = \frac{E_{i,j}^i}{E_j^i}, \quad (2)$$

Where $E_{i,j}^i$ is the energy fed to the grid by utility i during the j -th period and E_j^i is the total energy injected into the grid by the community during the period j .

It is also defined $ip_{i,j}$, i.e. the percentage of participation realised for energy inputs by prosumers only as:

$$ip_{i,j} = \begin{cases} \frac{E_{i,j}^{ip}}{E_j^{ip}} & \text{if user is prosumer} \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

Where E_j^{ip} is the total energy injected into the grid by the prosumers of the community during the j -th period.

It is technically possible to compute $i_{i,j}$, $w_{i,j}$ and $ip_{i,j}$ on a different time frames depending on how often the community decides to redistribute incentives. For a matter of simplicity in section 3 the redistribution is performed with the coefficients computed over a year calculated as average over the monthly values. As a direct consequence, the incentive gained over an entire year will be redistributed.

In Fig. 1 is reported the redistribution tree. It has maximum depth equal to five, with a total of nine nodes and five leaves. Using the same redistribution schema and metrics is possible to compute a reallocation on different time frames by modifying the dimensions of the vectors and matrixes that will be introduced later.

Each node (category) is associated with a quantity; said quantity has to be redistributed among users if the node is a leaf node otherwise it has to be apportioned in later shares. Each arc is associated with a percentage value that, consequently, determines the quantity of the next node.

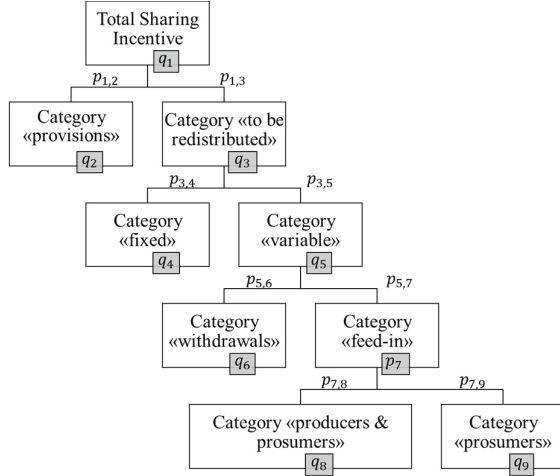


Figure 1: benefits redistribution tree.

Some prior definitions are required: let f the index that identifies the parent nodes and l the index that identifies the leaf nodes, so that $f \in F$ and $l \in L$ where F and L are respectively the set of parent nodes and leaves nodes. It is possible to define:

$$0 \leq p_{f,i} \leq 1, \quad (4)$$

$$\sum_i p_{f,i} \leq 1, \quad \forall f \in F, \quad (5)$$

$$q_i = q_{f,i} p_{f,i}, \quad (6)$$

$$\sum_l q_l = q_1, \quad (7)$$

Where q_i is the amount at the i -th node, q_{fi} is the quantity at the parent of the i -th node, $p_{f,i}$ is the weight of the arch that connects the i -th node to its parent node and Equation (4) set the domain of the arcs' weight; Equation (5) imposes that the sum of the weight that start from the same parent node cannot exceed the unitary value (i.e. avoids to redistribute more than the maximum quantity available at a parent node). It is worth noticing that it is an inequality, not an equality. Equation (6) defines how to compute the quantity at the i -th node. Equation (7) imposes that the sum of the amounts at the leaves correspond to the starting quantity; such calculation can be used to verify the correct distribution of the total amount.

The categories of redistribution are further discussed, as well as the mode in which the classes of users compete to each category. At the root node is allocated the total amount that has to be redistributed. It outputs two arcs to "reservoirs" and "to be redistributed" categories. Depending on internal community decisions, it may decide to set aside resources for the implementation of other projects. The reservoirs category is intended for such purpose. Note that reservoirs are not redistributed among users. This will come in handy later. "Category to be redistributed" holds the total amount that is supposed to be assigned to the members. This category forks into the "Fixed" category and into the "Variable" category.

The variable amount is redistributed to the users based on their withdrawal and feed-in performance as defined in Equation (1) and (2), while the fixed category is participated by each user equally (i.e., if there are 4 users, each of them will be allocated 25% of the amount in the category). There are two categories of users that can participate in feed-in: producers and prosumers. The incentive allocated to those who feed energy into the grid has been divided into a category for producers and another one for prosumers to achieve an optimal balance. With the presented categories all classes of users can benefit from a fixed incentive fee just by joining the community. Likewise, each user class participating in the "Withdrawals" category will be remunerated in proportion to the percentage of energy drawn from the grid during the corresponding period ($w_{i,j}$). The amount in the "Feed-in" category forks into two categories: one in which the assigned amount is redistributed among or producers and prosumers and another one for prosumers only. The prosumers' participation in the latter is calculated on the total energy fed in by the prosumers only as defined in Equation (3).

The creation of two separate categories between producers and prosumers is designed because it is believed that if large producers are present, they could monopolize the incentive for members who also feed energy into the grid, preventing prosumers from obtaining the expected benefits; therefore, in this way, dominance actions by producers can be prevented.

2.2 Mathematical formulation of the problem

In Section 3, the proposed problem will be solved by optimising several objective functions. To this end, the problem has been modelled with a free and open-source constraint modelling language [14] that allows users to write models in a way that approximates a mathematical formulation of the problem and then solve them.

Note that the incentive that is redistributed among members is the one in the category "to be redistributed". Hence, the amount in reservoirs can be calculated as the total amount minus the amount to be redistributed (q_3). This allows for a reduction in the dimensions of the system.

It is possible to define the mathematical system for calculating the total incentive for the i -th user as:

$$u = Aq \quad (8)$$

Where u is a vector of length equal to the number of users whose elements identify the total incentive obtained by the i -th member. Its elements are calculated as the product of the matrix A by the vector q , where q represents the quantities attributed to each leaf node except the "reservoirs" node. For ease of notation, we will denote this subset of leaf nodes as L_2 which is therefore of length equal to the number of elements in the set L minus 1. The matrix A , on the other hand, is composed of the withdrawal and input percentages of each user for each category, except the reservoirs category. Its elements will be identified by the letter a_{il} . Each column corresponds to a category, whilst the rows identify the user. The amount in the category "to be redistributed" can be calculated as follows:

$$I_{Alloc} = \sum_{l \in L_2} \sum_{i \in I} a_{il} q_l, \quad (9)$$

With such formulation, in considerations of the previously introduced constraints, it is possible to make the model more complex by introducing more constraints that force the problem to a lower space search. To facilitate the understanding of the model, a case study is presented in which the scenario is further complicated.

3 Case Study

A feasible distribution of incentives is to be found to satisfy a number of constraints, in particular a minimum percentage of cost reduction for users will be introduced. It is desired to find the ideal apportionment such that the incentive received by every user plus the corresponding RID, is greater than a certain percentage of the costs it incurs. To accomplish this goal, it is required a vector $\mathbf{r}$ of length equal to the number of users which elements corresponds to the RID gained by the i -th user and a vector $\mathbf{b}$ of same length containing the costs incurred by each user that are to be reduced. These could be a different amount for each user corresponding, for instance, to an instalment or the energy bill. Another vector is required, namely $\mathbf{z}$, that represents the percentage of the costs incurred by every user that is to be reduced.

To establish the amounts to be redistributed in each category, it is necessary to have data concerning an energy community. To this end, a REC located in the centre of Italy was simulated for the calendar year 2021. Utilities' hourly load curves have been simulated with [15][16], while for the production curve of the PVs the PVGIS tool has been employed [17].

Five users have been defined according to the representation in Figure 2 in which a cluster of users connected to the closer MV/LV substation have been taken into account. Other neighbouring clients not participating in the REC were not shown in the example figure.

The following members of the REC have been considered:

- User 1: 7 kWp PV plant grid connected. The self-consumption is realised by the inverter and to this end is considered negligible.
- User 2: A residential family with one person working from home. In this case the couple owns a 3 kWp PV plant.
- User 3: a working couple. Both persons working outside. No PV equipped.
- User 4: A family three members with one parent working from home. The family also is the owner of a 3 kWp PV plant.
- User 5: A family with a retired couple over 64 years of age.

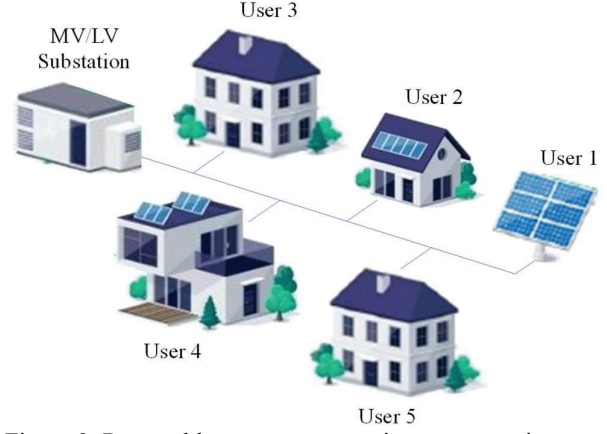


Figure 2: Renewable energy community representation.

The aim is to redistribute the total sharing incentive accumulated over the period of a year among users. Relatively to w_{ij} , i_{ij} and ip_{ij} the j index is lost since only one period is being considered (an entire year). This helps with the simplicity of the notation.

In Table 1 is reported a summary of the community members. Table 1 Users comprised in the simulated REC

User	Annual Consumption (kWh)	PV size (kWp)
User 1	150.00	7
User 2	3,309.01	3
User 3	3,159.05	0
User 4	4,727.23	3
User 5	2,351.82	0

Thanks to the simulated data, it has been possible to compute the shared energy on an hourly basis. The revenues generated by the RID have been calculated with the aid of average monthly prices that have been corresponded by the authority [18]. For each utility, the withdrawal and input percentages contained in the $\mathbf{A}$ matrix were calculated. As previously mentioned, the total incentive to be redistributed is the one derived from the shared energy. Note that q_9 is redistributed among prosumers according to the portion of energy fed into the grid out of the total energy fed in by all prosumers (Equation (3)), not by the overall community which could comprise also the producers. The total amount to be redistributes is 309,45€; in Table 2 are reported in detailed statistics relative to the users.

Table 2 Users statistics

User	w_i (%)	i_i (%)	ip_i (%)	r_i (€)	b_i (€)
User 1	0	68.5	0	465.0	1,344.0
User 2	17.7	16.8	53.7	117.9	446.6
User 3	30.8	0	0	0	764.8
User 4	28.6	14.7	46.3	102.5	727.51
User 5	22.9	0	0	0	564.34

Note one more time that in this case, w_i , i_i and ip_i are the averages of the percentages calculated on a monthly time basis.

As previously mentioned, the aim is to reduce a percentage of the costs incurred by a user. In this respect, thanks to the model created and the flexibility of the resolution software, it is possible to specify a unique minimum value for all users in a general way, or a customised value for all users.

So far, the general constraints have been set out, but the constraint for the abatement of costs for the individual user has not yet been imposed. This can be expressed as:

$$\sum_{l \in L_2} a_{il} q_l + r_i \geq b_i z_i, \quad \forall i \in I, \quad (10)$$

Where a_{il} is the element in position (i, l) of the matrix A and identifies the percentage of participation in the l -th category by the i -th user; q_l is the share at the l -th leaf node, whilst r_i , b_i and z_i are respectively the RID income, the incurred costs that are to be lowered and the percentage of minimum desired reduction for the i -th user.

It has to be underlined that the amount of incentive that is not redistributed among the members of the community can be calculated as the total incentive available minus the expression of Equation (9).

A simulation of the incentive redistribution among the members of the community is then carried out. Is made the assumption that the producer faces a monthly instalment of 112€, for a total of 1344€ per year. The objective is to lower this cost for the producer, while for the consumer and prosumer users in the community, the energy cost in the bill is considered. These data are reported in Table 2 as well.

The objective is to define the best allocation of amounts in the elements of the vector q under certain conditions (constraints). Note the fact that in Equation (10) the RID revenues are also considered in order to reach the cost reduction target. In this configuration, the minimum cost reduction guaranteed by the RID is 34.61% for user 1, 26.42% for user 2 and 14.09% for user 4. Five simulations were carried out with different constraints on the elements of the array q , as well as different objective functions. A summary of these is provided in Table 3.

As it can be observed, the constraints imposed on z_i concern its minimum value. Regarding the objective function, it varies and provides different configurations depending on the specifications. In simulation 1, the aim is to maximise the incentive redistributed to users (no quantity is allocated to the “reservoirs” category). Simulations two and three are equivalent regarding the objective function, which is simply expressed in two different forms. In simulation 2 the aim is to minimise the incentive distributed to the users, but to maintaining a minimum cost reduction equal to 10%, while this value is lowered in simulation 3 to 5%. In simulation 4, it is desired to maximise the cost reduction for prosumers (which are identified by indexes 2 and 4). The last simulation maximises the incentive received by the producer.

Table 3 Simulations: constraints on z_i and objective function

Simulation	Constraint on z_i	Objective function
1	$z_i \geq 0.10, \forall i \in I$	$\max(I_{Alloc})$
2	$z_i \geq 0.10, \forall i \in I$	$\max(I_{Alloc} - q_1)$
3	$z_i \geq 0.05, \forall i \in I$	$\min(I_{Alloc})$
4	$z_i \geq 0.10, \forall i \in I$	$\max(z_2 + z_4)$
5	$z_i \geq 0.08, \forall i \in I$	$\max(z_1)$

In table 4 and table 5 are reported respectively the values of vector q and vector z obtained during the simulations.

Table 4 Amount in each category (€) of L_2 for each simulation

Simulation	q_2	q_4	q_6	q_8	q_9
Sim. 1	0	0	309.45	0	0
Sim. 2	61.14	0	248.31	0	0
Sim. 3	185.3	0	124.15	0	0
Sim. 4	0	0	248.31	0	61.14
Sim. 5	0	0	198.66	110.79	0

Table 5 Reduction of costs (%) for each user in each simulation

Simulation	u_1	u_2	u_3	u_4	u_5
Sim. 1	34.61	37.44	12.93	25.60	12.96
Sim. 2	35.52	38.8	11.97	25.01	12.57
Sim. 3	34.61	31.23	5.18	18.71	5.20
Sim. 4	34.61	43.44	10.37	27.20	10.40
Sim. 5	40.26	38.37	8.27	23.71	8.29

As can be seen, in this simplified case, the category “Withdrawals” is sufficient to guarantee the achievement of the imposed targets. In the case of simulation 1, in which the entire incentive is redistributed among users, the cost reduction for consumer users (indexes 3 and 5) is maximum. However, in the case where the redistributed incentive is to be minimised, with the statistics of this case study, no quantity is allocated to categories 8 and 9 which are instead used in the case in which the incentive for producers and prosumers is to be maximised, respectively. The simulations also show that users with installations (producers and prosumers) benefit from higher cost reductions due to the valorisation of energy fed into the grid in addition to that obtained from self-consumption.

The chosen example shows how each redistribution category can be used when needed, while the use of factual statistics on energy withdrawals and injections into the grid allow redistribution on a meritocratic principle.

4 Conclusions and future developments

Energy communities are entities that allow users to obtain economic as well as environmental benefits. This gives rise to a critical aspect to be investigated in the management of

relations between its members: the repartition of revenues that are attributed to the entity responsible for the community as an incentive for the electricity production system associated with the energy community. The present work is set in this context, and develops a methodology for the meritocratic redistribution of economic benefits among the members of the energy community.

The methodology is based on the idea of identifying the different classes of users that are part of the energy community and allows a certain level of flexibility for the distribution of economic benefits among the members of the community without introducing implementation difficulties. In addition to the definition of user categories, meritocratic participation criteria realized on the percentage of energy withdrawn from and injected into the grid are defined, allowing for an optimal solution for the redistribution of incentives.

The effectiveness of the methodology is presented with the help of a case study that shows that the proposed approach is a valid option and that can provide different solutions to satisfy different objectives. The developed distribution scheme, including its constraints, could be supplemented by further data on the investment parameters related to the construction of the production plants. Future developments could concern the definition of a dedicated algorithm to overcome the use of the optimization software used in this study, as well as the expansion of the model with additional parameters that consider a broader community context or redistribution metrics based on other KPIs that consider other beneficial contributions to the energy community.

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